Rohan Alexander

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Academic appointments

University of Toronto
- Assistant Professor, Information (51%) & Statistical Sciences (49%) — Jul 2020 – *current*
  - Parental leave: Jun–Aug 2021.
  - COVID-19 childcare: 0.5 FTE Mar–Dec 2020; 0.25 FTE Jan–May 2021 and Sep–Dec 2021.
- Associate Director, Canadian Statistical Sciences Institute (CANSSI) Ontario — Sep 2021 – *current*
- Faculty Affiliate, Schwartz Reisman Institute for Technology and Society — Aug 2020 – *current*
- Director, Technical Skills Curriculum & Instruction, Data Sciences Institute — Jul 2021 – *current*
- Senior Fellow, Massey College — Sep 2021 – Sep 2024
- Co-lead, Data Sciences Institute Thematic Program in Reproducibility — Nov 2021 – Oct 2024
- Post-doctoral Fellow, Information, supervised by Kelly Lyons — Jul 2019 – Jul 2020

Education

Australian National University
- PhD (Economics) — Feb 2015 – Dec 2019
  - Dissertation title: *Essays on Applied Historical Labour Economics*.
  - Committee: John Tang (chair), Martine Mariotti, Tim Hatton, and Zach Ward.
- M. Economics — Feb 2012 – Nov 2012

University of Queensland
- B. Economics (Hon IIa) — Feb 2004 – Nov 2008
  - Honours thesis: *On Common Knowledge and Nash Equilibria*.

Book

- **Alexander, Rohan**, 2023, *Telling Stories with Data*, Chapman and Hall/CRC (second edition under contract for 2027 delivery).

Edited books

- **Alexander, Rohan** (editor), 2026, *Quantitative Social Science in the Age of AI*, accepted for publication by Chapman and Hall/CRC.
- Kennedy, Lauren, **Rohan Alexander**, Yajuan Si, and Andrew Gelman (editors), *Multilevel Regression and Poststratification: A Practical Guide and New Developments*, (under contract with Cambridge University Press).

Peer-reviewed publications

My work is interdisciplinary, and different disciplines have different norms for authorship and publication venue. I generally use the science approach of senior-last authorship, and almost always prioritize students for first-author. Two exceptions are Alexander and Ward, 2018, and Alexander and Hatton, 2025, which follow the economics approach of alphabetical authorship.

- Hughes, Evelyn and **Rohan Alexander**, 2026, "Evaluating AI Performance on End-to-End Data Science Projects", accepted *PLOS One*.

- Alexander, Monica, **Rohan Alexander**, and Leslie Root, 2026, "Decomposition of Black and White Infant Mortality Rates", *JAMA Network Open*, doi:10.1001/jamanetworkopen.2026.35321.
- **Alexander, Rohan**, Luca Carnegie, Nathalie Moon, 2026, "Self-reported LLM usage and outcomes on a data science project: Evidence from two undergraduate courses", *Journal of Statistics and Data Science Education*, doi:10.1080/26939169.2026.2708860.
- Brodeur, Abel, David Valenta, Alexandru Marcoci, Juan P. Aparicio, Derek Mikola, Bruno Barbarioli, **Rohan Alexander**, ..., 2026, "Comparing Human-Only, AI-Assisted, and AI-Led Teams on Assessing Research Reproducibility in Quantitative Social Science", *Proceedings of the National Academy of Sciences (PNAS)*, doi:10.1073/pnas.2524747123.
- Katz, Lindsay, Inessa De Angelis, and **Rohan Alexander**, 2026, "Do Women Politicians Face More Interruptions? An Analysis of Interjections in the Australian Parliamentary Debates (1998–2025)", *Australian Journal of Political Science*, doi:10.1080/10361146.2026.2642819.
- Lopez-Miranda, Ana Elisa, Tiffany Timbers, and **Rohan Alexander**, 2026, "Prompting the Professoriate: A Qualitative Study of Instructor Perspectives on LLMs in Data Science Education", *Harvard Data Science Review*, 8(1), doi:10.1162/99608f92.c8780a45.
- De Angelis, Inessa, and **Rohan Alexander**, 2026, "What are Canadian Members of Parliament doing on Bluesky?", *Canadian Journal of Political Science*, doi:10.1017/S0008423925100619.
- **Alexander, Rohan** and Timothy Hatton, 2025, "The making of a nation: Who voted for the Australian Federation?", *Asia-Pacific Economic History Review*, doi:10.1111/aehr.70008.
  - *2025 Sir Timothy Coghlan award for best paper in the Asia-Pacific Economic History Review.*
- Katz, Lindsay, and **Rohan Alexander**, 2023, "Digitization of the Australian Parliamentary Debates, 1998–2022", *Scientific Data*, 10, 567, doi:10.1038/s41597-023-02464-w.
- Collins, Annie, and **Rohan Alexander**, 2022, "Reproducibility of COVID-19 pre-prints", *Scientometrics*, 4 Jul, doi:10.1007/s11192-022-04418-2.
- **Alexander, Rohan**, and Kelly Lyons, 2020, "Barriers to Service Innovation Using Data Science", in *Advances in the Human Side of Service Engineering*, J. Spohrer and C. Leitner (Eds.): AHFE 2020, AISC 1208.
- **Alexander, Rohan**, and Zach Ward, 2018, "Age At Arrival and Assimilation In The Age Of Mass Migration", *The Journal of Economic History*, 78(3), pp. 904–937, doi:10.1017/S0022050718000335.

Other publications

- Ferguson, Sharon, Paula Akemi Aoyagui, **Rohan Alexander**, Anastasia Kuzminykh, 2022, "A Thematic Comparison of Human and AI Explanations of Sexism Assessment", *Human Centered AI Workshop at NeurIPS*.
- Horton, Nicholas, **Rohan Alexander**, Aneta Piekut, and Colin Rundel, 2022, "Editorial: The Growing Importance of Reproducibility and Responsible Workflow in the Data Science and Statistics Curriculum", *Journal of Statistics and Data Science Education*, doi:10.1080/26939169.2022.2141001.

Under submission

- Cowan, Michael, Martin Allen, Zane Schwartz, and **Rohan Alexander**, 2026, "Canadian Federal Government Procurement and Election Timing (2004–2025)", R&R at *Canadian Public Administration*.
- **Alexander, Rohan** and Annie Collins, 2026, "Simulating Gun Control Attitudes After the 2025 Bondi Beach Shooting Using Persona-Conditioned LLMs", R&R at *Australian Journal of Political Science*.
- Heath, Oscar and **Rohan Alexander**, 2026, "Correcting Mode Collapse in Silicon Sampling with Semantic Similarity Rating", reject and resubmit at *Political Analysis*, pre-print: arXiv:2607.28550.
- De Angelis, Inessa and **Rohan Alexander**, 2026, "New Platform, Old Strategies? Electioneering on Bluesky in the 2025 Canadian Federal Election", under review at *Canadian Journal of Political Science*.
- Cummins-Mburu, Benedict and **Rohan Alexander**, 2026, "Characterizing Canadian Out-of-district Political Donations from 2015 to 2024", under review at *Canadian Journal of Political Science*, pre-print: SocArXiv.

- Garcia Mejia, Mariana and **Rohan Alexander**, 2026, "If it bleeds, it leads? Evaluating whether LLMs can identify what is newsworthy", under review at *Computational Communication Research*, pre-print: doi:10.31235/osf.io/y6nqj_v1.
- Cui, Jiaxin (Allyson) and **Rohan Alexander**, 2026, "Same Prompt, Different Outcomes: Evaluating the Reproducibility of Data Analysis by LLMs", submitted to *Harvard Data Science Review*, pre-print: arXiv:2602.14349.
- Cummins-Mburu, Benedict, Mariana Garcia Mejia, Oscar Heath, Sabrina Kreyzerman, Ellie Murray, Arusan Surendiran and **Rohan Alexander**, 2026, "`GardenBench`: A lightweight, daily evaluation of LLM capabilities", pre-print: doi:10.5281/zenodo.22924865.
- Murray, Ellie and **Rohan Alexander**, 2026, "How analyst choices affect silicon samples from the 2025 CES", pre-print: SocArXiv.
- Aparicio, Juan P., David Valenta, Derek Mikola, Ghina Abdul Baki, Bruno Barbarioli, Luna Fazio, Lenka Fiala, Rohan Alexander, ..., 2026, "Generative AI boosts productivity across the expertise ladder".

Pre-prints

- **Alexander, Rohan**, Lindsay Katz, Callandra Moore, Zane Schwartz, 2023, "Evaluating the Decency and Consistency of Data Validation Tests Generated by LLMs", arXiv:2310.01402.
- Chiu, Ke-Li, Annie Collins, and **Rohan Alexander**, 2022, "Detecting Hate Speech with GPT-3", *arXiv*, 24 Mar, arXiv:2103.12407.
- **Alexander, Rohan**, and Monica Alexander, 2021, "The Increased Effect of Elections and Changing Prime Ministers on Topics Discussed in the Australian Federal Parliament between 1901 and 2018", *arXiv*, 17 Nov, arXiv:2111.09299.
- Chiu, Ke-Li, and **Rohan Alexander**, 2021, "On consistency scores in text data with an implementation in R", *arXiv*, 13 Jan, arXiv:2101.05225.
- Hodgetts, Paul A., and **Rohan Alexander**, 2020, "cesR: An R package for the Canadian Election Study", *SocArXiv*, 3 Sep, SocArXiv.

R packages

- **Alexander, Rohan**, and Paul A. Hodgetts, 2021, `AustralianPoliticians`.
- Hodgetts, Paul A., and **Rohan Alexander**, 2021, `cesR`.
- **Alexander, Rohan**, and A Mahfouz, 2021, `heapsofpapers`.

Awards

- 2025 Sir Timothy Coghlan award for best paper in the *Asia-Pacific Economic History Review*.
- Honourable Mention, 2021 Master of Information Student Council Instructor Awards.

Grants

PI means principal investigator. All amounts are Canadian dollars unless otherwise mentioned.

- Pangram, $500 API credits, (May 2026). "Evaluating AI use by Canadian politicians". Awarded to: **Rohan Alexander**.
- Anthropic, $500 API credits, (Apr 2026). "The Synthetic Voter Project". Awarded to: **Rohan Alexander**.
- Cohere Labs, $1,000 API credits, (Sep 2025). "Can LLMs Perform Data Analysis?". Awarded to: **Rohan Alexander**.

- SSHRC Partnership Grant–Stage 2, $2,499,979, (Apr 2025 – Mar 2032). "The Political Accountability, Transparency, and Representation Oversight Network". Director: Christopher Cochrane; Co-director: **Rohan Alexander**; Co-investigators: Angela Misri, April Lindgren, Harold Jansen, Holly A. Garnett, Jacob Robbins-Kanter, Laks V.S. Lakshmanan, Lisa M. Birch, Lisa Young, Michelle Alexopoulos, Monica Alexander, Nicole S. Blanchett, Robert Schertzer, Sara J. Bannerman, Semra Sevi, Shurui Zhou, Sophie Borwein, Tiffany A. Timbers, Vincent Arel-Bundock, Yannick Dufresne, Zachary T. Taylor, and Zane Schwartz.
- NSERC Alliance Grant and Mitacs Accelerate Internships, $718,129 (NSERC) and $800,000 (Mitacs), "Novel Statistical and Machine Learning Algorithms for Analysis of Canadian Political Donations and Lobbying Data", Feb 2025 – Dec 2029. PI: **Rohan Alexander**; co-PI: Laks Lakshmanan; collaborators: Shurui Zhou, Jesse Gronsbell.
- University of Toronto, Data Sciences Institute, Catalyst Grant, $100,000, "Automated LLM-based Pipelines to Improve Code Quality in the Social Sciences", Feb 2025 – Jan 2026. PI: **Rohan Alexander**; co-PIs: Monica Alexander and Chris Maddison.
- Mitacs Business Strategy Internship, $150,000 (delivered through paid internships), "Open by Default", Jan 2025 – Dec 2025. Supervisor: Christopher Cochrane; co-supervisor: **Rohan Alexander**.
- SSHRC Partnership Grant–Stage 1, $19,982, (May 2024 – Sep 2024) "The Political Accountability, Transparency, and Representation Oversight Network". Director: Christopher Cochrane; co-director: **Rohan Alexander**.
- SSHRC Partnership Development Grant, $172,511, (Mar 2024 – Mar 2026) "Transparency in Governance: Enhancing Data Integrity for Analyzing Conflicts of Interest and Lobbying in Canadian Politics". Director: **Rohan Alexander**; co-director: Zane Schwartz; co-investigators: Holly Ann Garnett, Semra Sevi; collaborators: April Lindgren, Chris Cochrane, Jacob Robbins-Kanter, Randy Besco.
- Open Philanthropy, $174,112 USD (~$235,342 CAD, as at 26 Feb 2024), (Apr 2024 – Jun 2025) "Benchmarking LLM agents on real-world tasks: Reproducibility". PI: Abel Brodeur; co-PIs: Alexandru Marcoci, and **Rohan Alexander**.
- University of Toronto, Data Sciences Institute, Seed Funding for Methodologists Grant, $10,000, (Feb 2024 – Jul 2024), "A methodological approach to automatically detect and improve issues in data science code with LLMs". PI: **Rohan Alexander**.
- University of Toronto, Data Sciences Institute, Computational & Quantitative Social Sciences Grant, $10,000, (Jan 2024 – Dec 2024), "Harmonizing Algorithms and Culture: Reconstructing Spotify's Danceability Metric to Explore the Algorithmic Rendering of Musical Style and Taste". PI: **Rohan Alexander**.
- SSHRC Partnership Engage Grant, $24,938, "Using machine learning to resolve data challenges to improve our understanding of Canadian lobbying activity", (Sep 2023 – Aug 2025). Partner organization: Investigative Journalism Foundation of Canada. PI: **Rohan Alexander**.
- Mitacs Accelerate, $426,666 (Mitacs: $235K, IJF: $192K) (Mitacs funding delivered through paid internships), "Follow the Money: Tracking the Influence of Lobbyists in Canada", (Jul 2023 – Jun 2026). Partner organization: Investigative Journalism Foundation of Canada. Supervisor: **Rohan Alexander**; co-supervisors: Christopher Cochrane, Raymond Ng.
- University of Toronto, Data Sciences Institute, Catalyst Grant, $100,000, "Providing data to improve representation by public officials", (Feb 2023 – Jan 2024). Named PI: Peter Loewen; co-PIs: **Rohan Alexander**, Aya Mitani, and Elena Tuzhilina.
- Canadian Statistical Sciences Institute (CANSSI) Ontario, Data Access Grant, $5,000, "Developing a large-scale lobbying and political donations dashboard to foster access to and interdisciplinary research on democratic accountability and political donations", (Jan 2023). Co-PIs: **Rohan Alexander** and Chris Cochrane.
- Australian Research Council (ARC), Discovery Projects, $287,146 AUD, "Australian Parliamentary Speech: How Deliberative? How Representative?", (Jan 2023 – Dec 2025). Funding: Chief Investigators: Marija Taflaga and Keith Dowding; partner investigators: Kenneth Benoit and **Rohan Alexander**.
- University of Toronto, Data Sciences Institute, Data Access Grant, $9,993, "Developing a large-scale lobbying and political donations dashboard to foster access to and interdisciplinary research on democratic accountability and political donations", (Nov 2022 – Nov 2023). Co-PIs: **Rohan Alexander** and Chris Cochrane.

- SSHRC Connection Grant, $50,000, "Follow the money: Mobilizing Canadian political donations data", (Oct 2022 – Sep 2024), PI: **Rohan Alexander**; co-investigators: Jacob Robbins-Kanter, Wing Cheung Michael Wong, and Zane Schwartz.
- SSHRC Partnership Development Grant, $169,855, (Jun 2022 – Jun 2024), "Follow the Money: Donors, Donations, and Parties in Canadian Politics". PI: Christopher Cochrane; co-applicants: Harold Jansen, Jacob Robbins-Kanter, Michael Pereira, Randy Besco, **Rohan Alexander**, Michael Wing-Cheung Wong, and Zane Schwartz.
- Mitacs Accelerate, $48,000 (delivered through paid internships), "Follow the Money: Investigating Political Donations in Canada", (Jun 2022 – Nov 2022). Partner organization: Investigative Journalism Foundation of Canada. Co-PIs: **Rohan Alexander** and April Lindgren.
- University of Toronto, Data Sciences Institute, SUDS, $7,500, "Disparities in climate-induced health outcomes in the Greater Toronto Area" (May 2022 – Aug 2022). **Rohan Alexander** and Monica Alexander.
- University of Toronto, School of Cities, Urban Challenge Grant, $75,000, "Disparities in climate-induced health outcomes in the Greater Toronto Area", (Jan 2022 – Dec 2022). Co-PIs: **Rohan Alexander**, Monica Alexander, Karen Smith, and Micah Hewer.
- University of Toronto, Data Sciences Institute, Data Access Grant, $10,000, (Dec 2021 – Apr 2022), "Disparities in climate-induced health outcomes in the Greater Toronto Area". Co-PIs: **Rohan Alexander** and Monica Alexander.
- University of Toronto, Faculty of Information, SSHRC Institutional Grant, $3,500, (Nov 2021 – Mar 2022), "At what cost? Using Hansard data to understand the public policy that went 'missing' due to COVID-19". PI: **Rohan Alexander**.
- University of Toronto, Faculty of Arts & Science, PIEF, $10,000, (Sep 2021 – Mar 2022), "Developing the DoSS toolkit". Co-PIs: **Rohan Alexander** and Samantha-Jo Caetano.
- CANSSI Ontario, Workshop/Lecture (COWL) Program, $1,000, "Reproducibility of COVID-19 pre-prints", (Feb 2021). PI: **Rohan Alexander**.
- CANSSI Ontario Multidisciplinary Doctoral (Mdoc) training program, $10,000 annually for five years, (Sep 2021 – Sep 2025). Co-PIs: **Rohan Alexander** and Monica Alexander.
- Mitacs Accelerate, $30,000, (Jun 2021 – Nov 2021), "Robust Taxonomy out of Receipt Item Labels", Partner organization: Sensibill. Co-PIs: **Rohan Alexander** and Michael Guerzhoy.
- University of Toronto, Faculty of Arts & Science, Pedagogical Innovation and Experimentation Fund, $15,000, (Sep 2020 – Mar 2021). "Developing an applied statistics toolkit". Co-PIs: **Rohan Alexander** and Samantha-Jo Caetano.
- University College London and University of Toronto, Strategic Partner Funds for AI collaborations, $25,000, (Jul 2020 – Dec 2020). "AI Governance: Building ethical frameworks to balance risk and innovation". Co-PIs: Elizabeth Lomas, Kelly Lyons, Jenny Bunn, and **Rohan Alexander**.

Teaching experience

Graduate

- Instructor, "STA2453: Data Science Methods, Collaborations, and Communication", Statistical Sciences — Fall 2026
- Instructor, "INF3104: Data Science Foundations", Information — Fall 2026
- Instructor, "STA4101: Topics in Statistical Data Science", Statistical Sciences — Fall 2025
- Instructor, "INF3104: Data Science Foundations", Information — Fall 2024
- Instructor, "INF3104: Data Science Foundations", Information — Winter 2023
- Instructor, "INF2178: Experimental Design for Data Science", Information — Winter 2021
- Instructor, "INF2178: Experimental Design for Data Science", Information — Winter 2020

Undergraduate

- Instructor, "STA304: Surveys, Sampling and Observational Data", Statistical Sciences — Fall 2024
- Instructor, "STA302: Methods of Data Analysis I", Statistical Sciences — Winter 2024
- Instructor, "INF312: Worlds Become Data", Information — Winter 2024
- Instructor, "INF312: Worlds Become Data", Information — Winter 2023
- Instructor, "INF312: Worlds Become Data", Information — Winter 2022
- Instructor, "STA304: Surveys, Sampling and Observational Data", Statistical Sciences — Winter 2022
- Instructor, "STA304: Surveys, Sampling and Observational Data", Statistical Sciences — Fall 2020

Supervisory experience

PhD

- Ana Elisa Lopez-Miranda, Statistical Sciences — Sep 2026 – *current*
- Inessa De Angelis, Information — Sep 2024 – *current*
- Ciara Zogheib, Information — Sep 2022 – *current*

Masters

- Xutong Jiang, Applied Computing (academic supervisor) — Apr 2024 – Dec 2024
  - Thesis: *Automated Financial Modelling from Contractual Definitions*
  - First placement: AI Research Developer, TAO Solutions Inc
- Inessa De Angelis, Information (thesis supervisor) — Sep 2023 – May 2024
  - Thesis: *Torrential Twitter: Climate Change, Female Politicians, and Harassment*
  - First placement: PhD student, University of Toronto
- Lorena Almaraz De La Garza, Information (academic supervisor) — Jun 2022 – Sep 2022
  - Research paper: *Machine Learning Inference for Demographic Features of Political Donors*
  - First placement: Senior AI/ML Model Steward Partner, Thomson Reuters
- Ciara Zogheib, Information (thesis supervisor) — Sep 2021 – Jun 2022
  - Thesis: *The "Paper and Pencil" Operations of Interdisciplinarity: Interdisciplinary Integration as Information Practice*
  - First placement: PhD student, University of Toronto
- Elisa Du, Applied Computing (academic supervisor) — Mar 2021 – Dec 2021
  - Thesis: *Enriched Understanding of Retail Receipts for Personalized Financial Insights*
  - First placement: Machine Learning Engineer, Sensibill

Undergraduate

- Connor Glossop, Engineering (senior thesis) — Sep 2022 – Apr 2023
  - Thesis: *Analyzing Political Polarization in Canada's Parliament*
  - First placement: Data Scientist, Intact Insurance
- Callie Moore, Engineering (senior thesis) — Sep 2021 – Apr 2022
  - Thesis: *Dyadic Representation in Canadian Parliament*
  - First placement: Data Analyst, Hospital for Sick Children
- Jeremy Hunt, Engineering (senior thesis) — Sep 2020 – Apr 2021
  - Thesis: *Study of Software Tool Usage in Academics*

Reading courses

- Oscar Heath, "Silicon Sampling for **in silico** Survey Design" — Fall 2026
- Amy Katz, "History of Statistics" — Fall 2026
- Ethan Sansom, "Type Systems and Data Validation for Statistical Computing" — Summer 2026
- Rosie Topp, "Generative AI and Social Science" — Summer 2026
- Andrew Gad, "Applied Causal Inference in Observational Data" — Summer 2026
- Ethan Sansom, "Statistical Software Design" — Fall 2025

- John Zhang, "Sports Analytics" Fall 2025
- Talia Fabregas, "Statistical Methods for Election Forecasting and Polling" Summer 2025
- Kiki He, "History of Statistics" Summer 2025
- Jack McKay, "Applications of Time Series Statistics" Fall 2024
- Inessa De Angelis, "Introduction to Statistical Modeling" Fall 2023
- Michaela Drouillard, "Harmonizing Algorithms and Culture" Fall 2023
- Christina Wei, "Natural Language Processing" Fall 2023
- Shaohan Chang, "Statistical Modeling" Fall 2023
- Ananya Jha, "Ethics in Data Science and AI" Winter 2023
- Luyang Shang, "Natural Language Processing" Fall 2022
- Christina Dinh Nguyen, "Natural Language Processing" Fall 2022
- Rod Mazloomi, "Natural Language Processing" Fall 2022
- Brian Diep, "Natural Language Processing" Fall 2022
- Thomas William Rosenthal, "A Machine Learning Exploration of Single-Origin Coffee" Winter 2021
- Thomas William Rosenthal, "Ethics and Data Science" Winter 2021
- Paul Hodgetts, "Ethics and Data Science" Winter 2021
- Ke-Li Chiu, "Ethics and Data Science" Winter 2021
- Asel Kushkeyeva, "Ethics and Data Science" Winter 2021
- Henry Rowan Shiffer, "Introduction to Statistical Modeling" Summer 2021
- Siqi Zheng, "Introduction to Statistical Modeling" Fall 2021
- Changhao Jiang, "Introduction to Statistical Modeling" Fall 2021
- Chengen Dong, "Introduction to Statistical Modeling" Fall 2021
- Haoluan Chen, "Natural Language Processing" Fall 2020

Service

Professional societies

- President-elect; President; Past-president, *Data Science and Analytics Section, Statistical Society of Canada* Jul 2023 – Jun 2026

Editorial

- Associate Editor, *Journal of Statistics and Data Science Education* Summer 2021 – Summer 2025

Refereeing

- 2026: *The R Journal*; *Communications AI & Computing*; *Chapman and Hall/CRC*; *Harvard Data Science Review*; *Political Science Research and Methods*; *Population and Development Review*; *UKRI Funding Service.*
- 2025: *The Journal of Economic History*; *Journal of Statistics and Data Science Education* (x2); *The American Statistician*; *Harvard Data Science Review*; *Canadian Journal of Political Science.*
- 2024: *Journal of Statistics and Data Science Education.*
- 2023: *Journal of Statistics and Data Science Education* (x3); *Social Sciences and Humanities Research Council of Canada*; *Archival Science*; *Harvard Data Science Review* (x2); *Demographic Research*; *PLOS Computational Biology*; *Canadian Journal of Political Science*; Mitacs Accelerate.
- 2022: *Chapman and Hall/CRC*; *Journal of Statistics and Data Science Education* (x3); *Political Analysis*; *Records Management Journal.*
- 2021: *Chapman and Hall/CRC* (x3); *Journal of Statistics and Data Science Education* (x2).
- 2020: *Canadian Journal of Political Science* (x2); *Political Analysis*; *Harvard Data Science Review*; *The Economic Record.*

Outreach

- Co-organizer, *Toronto Data Workshop* Sep 2019 – *current*
- Organizer, *2nd Colloquium on Applications of Large Language Models* 1 May 2026

- Local host, *Toronto Replication Games* 29 Apr 2026
- Judge, *Open Data Awards*, City of Toronto 13 Jan 2026
- Moderator, *Book Talk: "Digital Dreams Have Become Nightmares: What We Must Do"*, U of T Department of Computer Science and the Schwartz Reisman Institute for Technology and Society 10 Sep 2024
- Presenter, *Adventures In Science Mentorship Program*, U of T Faculty of Medicine 20 Mar 2024
- Organizer, *Florence Nightingale Day* 2 Mar 2024
- Presenter, *Generative AI's Issues and Opportunities*, St Andrew's "Heart of the City" 27 Sep 2023
- Presenter, *Data Science in Action Summer High School Camp*, Harvard Biostatistics 19 Jul 2023
- Organizer, *1st Colloquium on Applications of Large Language Models* 11 Apr 2023
- Guest lecture, *Climate justice and data science*, "INF1322H: Communities and Values" 7 Mar 2023
- Organizer, *3rd Toronto Workshop on Reproducibility* 23 Feb 2023
- Local host, *Toronto Replication Games* 22 Feb 2023
- Presenter, *Adventures In Science Mentorship Program*, U of T Faculty of Medicine 16 Feb 2023
- Organizer, *Florence Nightingale Day* 18 Feb 2023
- Organizer, *CANSSI Ontario Statistical Software Conference* 10 Nov 2022
- Organizer, *Florence Nightingale Day* 9 Apr 2022
- Organizer, *2nd Toronto Workshop on Reproducibility* 23–25 Feb 2022
- Presenter, *Data Science in Action Summer High School Camp*, Harvard Biostatistics 7 Jul 2021
- Organizer, *1st Toronto Workshop on Reproducibility* 25–26 Feb 2021
- Co-organizer, *MRP Working Group* Jan 2020 – Nov 2021

Departmental and University Committees

- Member, Graduate Committee on Standing, Information Summer 2025 – *current*
- Member, Faculty Advisory Council, School of Cities Aug 2024 – *current*
- Member, Graduate Committee, Statistical Sciences Jul 2020 – *current*
- Reviewer, Graduate Fellowships, Schwartz Reisman Institute Feb 2026
- Member, Data Sciences Speaker Series Committee, Data Sciences Institute Summer 2023 – Summer 2025
- Member, Search Committee, HCDS Assistant Professor, Teaching May 2025 – Nov 2025
- Reviewer, Urban Challenge Grants, School of Cities Mar 2025
- Reviewer, Graduate Fellowships, Schwartz Reisman Institute Feb 2025
- Coordinator, Applied Research and Education Seminar Series, Statistical Sciences Jul 2022 – Jun 2024
- Member, Undergraduate Advisory Group, Faculty of Information Jul 2023 – Jun 2024
- Member, Search Committee, ISD/HCDS Assistant Professor, Faculty of Information Aug 2023 – Apr 2024
- Co-organizer, *Statistical Sciences PhD Practice Job Talk Day* 14 Dec 2023
- Member, BI Recruitment and Admission Committee, Information Jul 2021 – Jun 2023
- Member, Advisory Committee, Dean Search, Faculty of Information Nov 2022 – May 2023
- Co-organizer, *Statistical Sciences PhD Practice Job Talk Day* 12 Dec 2022
- Chair, Data Sciences Speaker Series Committee, Data Sciences Institute Mar 2022 – Mar 2023
- Member, Search Committee, Dalla Lana School of Public Health, Assistant Professor—Data Science Methods for Population Health and Health Systems Jul 2022 – Mar 2023
- Member, Faculty of Arts and Sciences Computation and Data Science Education Working Group Sep 2021 – Jul 2022
- Member, Applied Research and Education Seminar Series, Statistical Sciences Jul 2020 – Jun 2022
- Advisor, *Statistical Sciences PhD Student Research Day* 25 May 2022
- Co-organizer, *Statistical Sciences PhD Practice Job Talk Day* 17 Dec 2021
- Member, U of T Library Technical Skills Outreach Project Jul 2021 – Jun 2022
- Member, Search Committee, Information, HCDS Teaching Stream Assistant Professor Nov 2021 – Apr 2022
- Member, Search Committee, Information, Learning Hub Director and Teaching Stream Assistant Professor Nov 2020 – Apr 2021
- Member, Awards Committee, Information Jul 2020 – Nov 2020

Thesis committees

- Member, Information, PhD Thesis Committee, Eshta Bhardwaj — Aug 2024 – *current*
  - Title: *Adopting a Data Curation Lens for Rigorous, Reflexive, and Responsible Machine Learning Data Practices*
- Member, Criminology, PhD Thesis Committee, Jamie Duncan — Jan 2023 – Jul 2026
  - Title: *The border as a platform*
- Member, Information, PhD Thesis Committee, Mirai Gendi — Jan 2022 – Aug 2026
  - Title: *From Forensic Narratives to Event Logs: Process-Mining Based Reconstruction and Defensible Forensic Process Models*

Exam committees

- Member, Statistical Sciences, PhD Final Oral Examination, Ziyi Liu — 14 Sep 2026
- Chair, Information, Masters Final Oral Examination, Rebecca Matthews — 31 Aug 2026
- Chair, Information, PhD Proposal Defence, Sara Al Hajj Ibrahim — 27 Aug 2026
- Member, Information, PhD Final Oral Examination, Mirai Gendi — 22 Jul 2026
- Member, Criminology, PhD Final Oral Examination, Jamie Duncan — 21 Jul 2026
- Member, Statistical Sciences, PhD Final Oral Examination, Ziyi Liu — 7 Jul 2026
- Member, Statistical Sciences, PhD Final Oral Examination, Liam Welsh — 24 Jun 2026
- Member, Statistical Sciences, Final Oral Examination, Tong Li — 29 May 2026
- Member, Statistical Sciences, PhD Final Oral Examination, Michael Moon — 1 Apr 2026
- Member, Statistical Sciences, PhD Departmental Oral Examination, Tong Li — 6 Mar 2026
- Chair, Information, Masters Final Oral Examination, Sophia Art — 18 Dec 2025
- Chair, School of Graduate Studies, Final Oral Examination, Alexandra Ballyk — 9 Jun 2025
- Chair, Statistical Sciences, Department Oral Examination, David Veitch — 3 Jun 2025
- Chair, Information, Masters Final Oral Examination, Shucen Lu — 3 Sep 2024
- Member, Information, PhD Qualifying Exam Committee, Eshta Bhardwaj — Fall 2023 – Winter 2024
- Member, Information, PhD Qualifying Exam Committee, Christina Wei — Fall 2023 – Winter 2024
- Member, Information, Thesis Proposal Defence, Mirai Gendi — Summer 2022 – Summer 2023
- Examiner, Statistical Sciences, PhD Applied Research Comprehensive, Fred Song — Fall 2023
- Chair, School of Graduate Studies, Final Oral Examination, Amanda Brijmohan — 30 Aug 2023
- Member, Statistical Sciences, PhD Final Oral Examination, Blair Bilodeau — 14 Apr 2023
- Chair, Statistical Sciences, PhD Oral Exam Committee, Michaël Lalancette — 14 Jul 2022
- Second Reader, Information, MI Thesis Defense, June Lim — 9 Sep 2021
- Member, Statistical Sciences, PhD Oral Exam Committee, Alex Stringer — 27 Apr 2021
- Member, Statistical Sciences, PhD Oral Exam Committee, Arvind Shrivats — 12 Feb 2021
- Member, Statistical Sciences, PhD Oral Exam Committee, Zhen Qin — 12 Jan 2021

Keynote

- *Telling Stories with Data in the AI Age*
  - 16 Jun 2026, 2026 Electronic Conference on Teaching Statistics (eCOTS), online.

Presentations

- *Iterated creation, grading, and revision of data science projects by language models*
  - (Poster) 3 Oct 2026, Empirical Methods in the Age of AI, Stanford.
- *Simulating Gun Control Attitudes After the 2025 Bondi Beach Shooting Using Persona-Conditioned LLMs*
  - 6 May 2026, MapleMeth, Toronto.
- (Chair) *Computational & Qualitative Methods I*
  - 25 April 2026, American Sociology Association, Midyear Methods Meeting, Toronto.

- *Benchmarking AI Performance on End-To-End Data Science Projects*
  - 26 Feb 2026, UMass Boston Stats Snack (online).
  - 17 Jan 2026, Southern Political Science Association Conference, New Orleans.
- (Panelist) *Science, Validation, and LLMs in Political Science*
  - 17 Jan 2026, Southern Political Science Association Conference, New Orleans.
- *Evaluating Analytical Variation in Large Language Models: A Many-Analysts Approach to Data Science Tasks*
  - 4 Dec 2025, Australian Society for Quantitative Political Science Conference, Melbourne.
- *Living in interesting times: Some opportunities and threats from AI in quantitative social sciences*
  - 3 Jul 2025, University of Sydney, Australia.
- *Getting started with Natural Language Processing*
  - 25 Jun 2025, University of Sydney, Australia.
- (Chair) *On Florence Nightingale Day*
  - 6 Aug 2024, Joint Statistical Meetings (JSM), Portland, Oregon.
- *Trustworthy Data Science and Self-Reported LLM Usage*
  - 4 Aug 2024, Joint Statistical Meetings (JSM), Portland, Oregon.
- *Reproducibility and code*
  - 3 Jun 2024, Statistical Society of Canada (SSC), Annual Meeting, Newfoundland.
- *Using LLMs to improve the trustworthiness of data science*
  - 8 May 2024, Theory of Data Science Retreat, (online).
- *AI and Data Science Education*
  - 6 May 2024, Panelist, Data Science Leadership Summit, Atlanta.
- *Automated testing of data science code with LLMs*
  - 1 May 2024, STATSTRO Workshop, Toronto.
- *Fixing Social Science by Integrating AI*
  - 26 Apr 2024, SRI Research Exchange & Social, Toronto.
- *Improving Reproducibility in Quantitative Social Sciences: Establishing a Simulation-Based Workflow Enhanced with Large Language Models*
  - 6 Mar 2024, Schwartz Reisman Institute for Technology and Society, Toronto.
- *Evaluating the Decency and Consistency of Data Validation Tests Generated by LLMs*
  - 7 Dec 2023, Australian Society for Quantitative Political Science, Brisbane.
  - 28 Nov 2023, Department of Statistical Sciences, Brown bag, Toronto.
  - 7 Nov 2023, Department of Mathematics & Statistics, University of Guelph.
- *Integrating Large Language Models (LLMs) into the Data Science Workflow*
  - 27 Oct 2023, Statistical Computing and Visualization (SCV) Committee, Statistics Society of Australia, (online).
- *From teaching to practice: Insights from the Toronto Reproducibility Conferences*
  - 8 Aug 2023, Joint Statistical Meetings (JSM), Toronto.
- *The Making of a Nation: Who Voted for Australian Federation?*
  - 23 Jun 2023, Australia before Federation, Centre for Economic History, Canberra.
- *Testing, testing, is this thing right?*
  - 27 Jan 2023, Quantitative Methods Cluster, University of Toronto.
  - 30 Nov 2022, FCT Abuja, Nigeria, UseR Group, (online).
  - 9 Nov 2022, Workshop in Computational & Quantitative Social Sciences, Toronto.
- *The growing importance of reproducibility and responsible workflow in the data science and statistics curriculum*
  - 13 Dec 2022, Panelist, CAUSE/JSDSE Webinar Series (online).
- *Reproducible workflows*
  - 2 Nov 2022, Biostatistics seminar series, Centre for Outcomes Research, McGill University Health Centre – Research Institute (online).
- *Rethinking data science*
  - 6 Oct 2022, Toronto Data Workshop, Toronto.

- *On the role of reproducibility in data science*
  - 4 Aug 2022, Stanford Data Science (online).
- *Integrating reproducibility into undergraduate data science teaching*
  - 14 Jul 2022, Australasian Teaching Economics Conference, (online).
  - 4 Jun 2022, Canadian Economics Association Conference, Carleton University, Ottawa.
- *On the double burden of disadvantage in data science*
  - 17 Jun 2022, We All Count, Talking Data Equity (online).
- *Reproducible Data Science Workflows in R*
  - 11 Jan 2022, Accra R User Ghana R Conference (online)
- *What data science can learn from the James Webb Space Telescope*
  - 17 Dec 2021, University of Toronto Statistics and MachIne LEarning (SMILE) Journal Club (online).
- *On the privilege of turning our world into data*
  - 16 Dec 2021, Young Irish Statisticians (online).
- *A super opinionated talk about data science, APIs, and grad school applications*
  - 18 Nov 2021, University of Toronto, Undergraduate Statistical Sciences Union Workshop (online).
- *Remarks on data science*
  - 19 Oct 2021, University of Toronto, Trinity College High Table (online).
- *Introduction to Andrew Gelman*
  - 17 Sep 2021, University of Toronto, Data Sciences Institute launch (online).
- *Introduction to R*
  - 22 Dec 2020, University of Toronto, Global Society for Genetics and Genome Biology (online).
- *Choosing the optimal set of co-variates to satisfy competing priorities in multi-level regression with post-stratification*
  - 6 May 2020, MRP Reading Group (online).
- *Forecasting Parliamentary Elections*
  - 10 Dec 2019, Monash University, Department of Econometrics & Business Statistics, Melbourne.
  - 9 Dec 2019, Australian Society for Quantitative Political Science Conference, Melbourne.
  - 2 Dec 2019, Australian National University, Cake for Comments, Canberra.
  - 20 Nov 2019, POP Politics Workshop, Canberra.
  - 17 Oct 2018, University of Toronto, Political Behaviour Group, Toronto.
- *A Word-Count Based Classifier of Politicians in the Australian Federal Parliament (1901–2018)*
  - 4 Sep 2019, European Consortium for Political Research, General Conference, Wroclaw.
  - 22 Jun 2019, European Political Science Association, Annual Conference, Belfast.
  - 18 Jun 2019, Quantitative Text Analysis Workshop, Dublin.
- *The Effect of Elections and Prime Ministers on Discussion in the Australian Federal Parliament (1901–2018)*
  - 11 Jan 2019, Political Methodology Specialist Group, Annual Conference, Warwick.
  - 11 Dec 2018, Max Planck Institute for Demographic Research, Rostock.
  - 22 Nov 2018, University of Toronto, Political Behaviour Group, Toronto.
  - 25 Oct 2018, Australian National University, Research School of Economics, Canberra.
  - 25 Oct 2018, Parliamentary Library, Parliament of Australia, Canberra.
  - 24 Oct 2018, Australian National University, School of Politics and International Relations, Canberra.
- *Exploring Australia's Hansard (1901–2017)*
  - (Poster) 7–9 Sep, 2018 Economic History Association Meeting, Montreal.
- *A Surnames-Based Analysis of Tasmanian Social Mobility (1803–2015)*
  - 20 May 2016, UC Berkeley, Economic History Lunch, Berkeley.
  - 2 May 2016, Australian National University, Research School of Economics, Canberra.